

LESSON 8-5 • SESSION 1 • PRACTICE • FORM B

Graphing Inequalities

Learning target – I can graph an inequality on a number line using an open or a closed dot. · 6.EE.B.8

REMEMBER

- An **open dot** means the boundary is NOT a solution ($<$ or $>$). A **closed dot** means it is ($\leq$ or $\geq$).
- Point the arrow toward the numbers that work: $>$ and $\geq$ point **right**; $<$ and $\leq$ point **left**.
- **At least** means $\geq$. **At most** means $\leq$. **Under** means $<$.

WORKED EXAMPLE

Graph $x < 6$ on a number line.**Step 1** The boundary number is 6.**Step 2** $<$ does not include 6, so draw an open dot at 6.**Step 3** Less than 6 means the arrow points left. Test: $5 < 6$ is true.**Answer** open dot at 6, arrow pointing left**WORDS** boundary point (*punto límite*) · closed circle (*círculo cerrado*) · open circle (*círculo abierto*)

PRACTICE

show your work · label units

1. Describe each graph: open or closed dot, and the arrow direction.

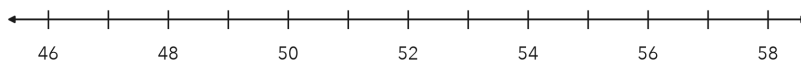
a. $k \leq -4$ _____

b. $n > 10$ _____

c. $p < -2$ _____

d. $x \geq 1$ _____

- 2.
- Roller Coaster.**
- A roller coaster lets a rider on only when the rider is at least 52 inches tall. Let
- h
- be the height in inches.

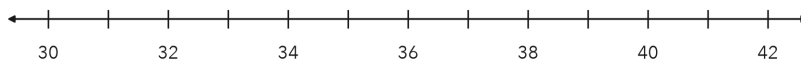


a. Write the inequality. _____

b. Graph it on the number line.

c. Name one value that works and one that does not. _____

- 3.
- Bike Rack.**
- A bike rack outside the gym holds at most 36 bikes. Write an inequality using
- b
- . Then graph it.

Hint "At most" means the boundary is allowed, so fill the dot.

Inequality _____

- 4.
- Pool Closed.**
- A pool closes when the air temperature drops to 65 degrees or lower. Write an inequality using
- t
- and describe its graph.

Hint "Or lower" means 65 itself counts.

Show your work.

Inequality _____

- 5.
- Lap Lane.**
- Swimmers must be at least 12 years old to use the lap lane. Write an inequality using
- a
- and describe its graph.

Hint "At least" means age 12 is allowed.

Show your work.

Inequality _____

6. Read a Graph. Use the graph below.

Hint An open dot leaves the boundary out; a closed dot keeps it.



a. Write the inequality, using x . _____

b. Is 2 a solution? Explain. _____

7. Canned Food Drive. A homeroom wins a breakfast when it collects at least 180 cans. Let c be the number of cans.

Hint Any total of 180 or more wins the breakfast.

a. Write the inequality. _____

b. Name two totals under your arrow. _____

8. Two Signs. A ride sign says riders must be at least 48 inches tall. A bridge sign says trucks must weigh under 10 tons.

a. Write an inequality for the first sign, using h . _____

b. Write an inequality for the second sign, using w . _____

c. Explain why one graph uses a closed dot and the other an open dot. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. Omar's Work. Omar graphed $x \leq 7$. He drew a closed dot at 7 and then drew his arrow pointing to the right. Explain his mistake and describe the correct graph.

TEST PRACTICE

choose the best answer

10. The graph of $x < 8$ has –

A a closed dot at 8, arrow left

B an open dot at 8, arrow right

C an open dot at 8, arrow left

D a closed dot at 8, arrow right

11. A closed dot at the boundary means the boundary value is –

A a solution

B not a solution

C negative

D zero

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

$x < 2$ C	$c \geq 180$ L	$b \geq 36$ Z	$h \geq 52$ C	$h > 52$ Q	$w < 10$ E	$w \leq 10$ J	$b \leq 36$ I	$x \leq 2$ X	$t \leq 65$ R
#2a		#3	#4	#6a	#7	#8b			

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____